

PL/SQL Practical Assignment

Complete Programs with Outputs (Questions 1 to 21)

1. Write A Program To Print "Welcome To PL/SQL Block"

```
SET SERVEROUTPUT ON;
BEGIN
    DBMS_OUTPUT.PUT_LINE('Welcome To PL/SQL Block');
END;
/
```

OUTPUT

Welcome To PL/SQL Block

2. Write A Program To Count Square

```
SET SERVEROUTPUT ON;
DECLARE
    num NUMBER := 5;
    sq NUMBER;
BEGIN
    sq := num * num;
    DBMS_OUTPUT.PUT_LINE('Square of ' || num || ' is: ' || sq);
END;
/
```

OUTPUT

Square of 5 is: 25

3. Write A Program To Find Simple Interest

```
SET SERVEROUTPUT ON;
DECLARE
  p NUMBER := 1000;
  r NUMBER := 5;
  t NUMBER := 2;
  si NUMBER;
BEGIN
  si := (p * r * t) / 100;
  DBMS_OUTPUT.PUT_LINE('Simple Interest is: ' || si);
END;
/
```

OUTPUT

```
Simple Interest is: 100
```

4. Write a program for global and local variable

```
SET SERVEROUTPUT ON;
DECLARE
  g_var NUMBER := 100; -- Global variable
BEGIN
  DBMS_OUTPUT.PUT_LINE('Outer Variable: ' || g_var);
  DECLARE
    l_var NUMBER := 50; -- Local variable
  BEGIN
    DBMS_OUTPUT.PUT_LINE('Inner Local Variable: ' || l_var);
    DBMS_OUTPUT.PUT_LINE('Accessing Global inside: ' || g_var);
  END;
END;
/
```

OUTPUT

```
Outer Variable: 100
Inner Local Variable: 50
Accessing Global inside: 100
```

5. Write a program to give a declare constant variable

```
SET SERVEROUTPUT ON;
DECLARE
    pi CONSTANT NUMBER := 3.14;
    radius NUMBER := 5;
    area NUMBER;
BEGIN
    area := pi * radius * radius;
    DBMS_OUTPUT.PUT_LINE('Area of Circle: ' || area);
END;
/
```

OUTPUT

```
Area of Circle: 78.5
```

6. Write a program to check ticket age wise with the use of IF-THEN-ELSIF-ELSE

```
SET SERVEROUTPUT ON;
DECLARE
    age NUMBER := 25;
BEGIN
    IF age < 5 THEN
        DBMS_OUTPUT.PUT_LINE('Ticket is Free');
    ELSIF age >= 5 AND age <= 18 THEN
        DBMS_OUTPUT.PUT_LINE('Child Ticket: $5');
    ELSIF age > 18 AND age <= 60 THEN
        DBMS_OUTPUT.PUT_LINE('Adult Ticket: $10');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Senior Citizen Ticket: $6');
    END IF;
END;
/
```

OUTPUT

```
Adult Ticket: $10
```

7. Write a program to input three no and find out maximum and minimum from it

```
SET SERVEROUTPUT ON;
DECLARE
  a NUMBER := 15;
  b NUMBER := 42;
  c NUMBER := 8;
  max_val NUMBER;
  min_val NUMBER;
BEGIN
  IF a >= b AND a >= c THEN max_val := a;
  ELSIF b >= a AND b >= c THEN max_val := b;
  ELSE max_val := c;
  END IF;

  IF a <= b AND a <= c THEN min_val := a;
  ELSIF b <= a AND b <= c THEN min_val := b;
  ELSE min_val := c;
  END IF;

  DBMS_OUTPUT.PUT_LINE('Maximum: ' || max_val);
  DBMS_OUTPUT.PUT_LINE('Minimum: ' || min_val);
END;
/
```

OUTPUT

```
Maximum: 42
Minimum: 8
```

8. Write a program to input rollno and three subject marks. find out total, percentage, result and grade

```
SET SERVEROUTPUT ON;
DECLARE
    rollno NUMBER := 101;
    m1 NUMBER := 75;
    m2 NUMBER := 80;
    m3 NUMBER := 85;
    total NUMBER;
    percentage NUMBER;
    grade VARCHAR2(15);
BEGIN
    total := m1 + m2 + m3;
    percentage := total / 3;

    IF percentage >= 75 THEN grade := 'Distinction';
    ELSIF percentage >= 60 THEN grade := 'First Class';
    ELSE grade := 'Pass';
    END IF;

    DBMS_OUTPUT.PUT_LINE('Roll No: ' || rollno);
    DBMS_OUTPUT.PUT_LINE('Total: ' || total);
    DBMS_OUTPUT.PUT_LINE('Percentage: ' || percentage || '%');
    DBMS_OUTPUT.PUT_LINE('Grade: ' || grade);
END;
/
```

OUTPUT

```
Roll No: 101
Total: 240
Percentage: 80%
Grade: Distinction
```

9. Write a program for the use of PL/SQL Case Statement

```
SET SERVEROUTPUT ON;
DECLARE
    day_num NUMBER := 3;
BEGIN
    CASE day_num
        WHEN 1 THEN DBMS_OUTPUT.PUT_LINE('Monday');
        WHEN 2 THEN DBMS_OUTPUT.PUT_LINE('Tuesday');
        WHEN 3 THEN DBMS_OUTPUT.PUT_LINE('Wednesday');
        ELSE DBMS_OUTPUT.PUT_LINE('Invalid Day');
    END CASE;
END;
/
```

OUTPUT

Wednesday

10. Write a program to print first 10 numbers using basic loop.. exit loop

```
SET SERVEROUTPUT ON;
DECLARE
    i NUMBER := 1;
BEGIN
    LOOP
        DBMS_OUTPUT.PUT_LINE(i);
        i := i + 1;
        EXIT WHEN i > 10;
    END LOOP;
END;
/
```

OUTPUT

1
2
3
4
5
6
7
8
9
10

11. Write a program to print first 10 numbers using while loop

```
SET SERVEROUTPUT ON;
DECLARE
    i NUMBER := 1;
BEGIN
    WHILE i <= 10 LOOP
        DBMS_OUTPUT.PUT_LINE(i);
        i := i + 1;
    END LOOP;
END;
/
```

OUTPUT

```
1
2
3
4
5
6
7
8
9
10
```

12. Write a program to print first 10 numbers using for loop

```
SET SERVEROUTPUT ON;
BEGIN
    FOR i IN 1..10 LOOP
        DBMS_OUTPUT.PUT_LINE(i);
    END LOOP;
END;
/
```

OUTPUT

```
1
2
3
4
5
6
7
8
9
10
```

13. Write a program to print first 1 to 10 numbers in reverse using for loop

```
SET SERVEROUTPUT ON;
BEGIN
  FOR i IN REVERSE 1..10 LOOP
    DBMS_OUTPUT.PUT_LINE(i);
  END LOOP;
END;
/
```

OUTPUT

```
10
9
8
7
6
5
4
3
2
1
```

14. Write a program to print first 10 prime number using while loop

```
SET SERVEROUTPUT ON;
DECLARE
    n NUMBER := 2;
    cnt NUMBER := 0;
    i NUMBER;
    is_prime BOOLEAN;
BEGIN
    WHILE cnt < 10 LOOP
        is_prime := TRUE;
        i := 2;
        WHILE i <= TRUNC(SQRT(n)) LOOP
            IF MOD(n, i) = 0 THEN
                is_prime := FALSE;
                EXIT;
            END IF;
            i := i + 1;
        END LOOP;
        IF is_prime THEN
            DBMS_OUTPUT.PUT_LINE(n);
            cnt := cnt + 1;
        END IF;
        n := n + 1;
    END LOOP;
END;
/
```

OUTPUT

```
2
3
5
7
11
13
17
19
23
29
```

15. Write a program to accept a string from user and display reverse of it

```
SET SERVEROUTPUT ON;
DECLARE
    str VARCHAR2(20) := 'HELLO';
    rev_str VARCHAR2(20);
BEGIN
    FOR i IN REVERSE 1..LENGTH(str) LOOP
        rev_str := rev_str || SUBSTR(str, i, 1);
    END LOOP;
    DBMS_OUTPUT.PUT_LINE('Reversed String: ' || rev_str);
END;
/
```

OUTPUT

Reversed String: OLLEH

16. Write a program to input a string and display its Pyramid

```
SET SERVEROUTPUT ON;
DECLARE
    str VARCHAR2(20) := 'JAVA';
BEGIN
    FOR i IN 1..LENGTH(str) LOOP
        DBMS_OUTPUT.PUT_LINE(SUBSTR(str, 1, i));
    END LOOP;
END;
/
```

OUTPUT

J
JA
JAV
JAVA

17. Write a program to print number pyramid

```
SET SERVEROUTPUT ON;
BEGIN
  FOR i IN 1..4 LOOP
    FOR j IN 1..i LOOP
      DBMS_OUTPUT.PUT(j || ' ');
    END LOOP;
    DBMS_OUTPUT.NEW_LINE;
  END LOOP;
END;
/
```

OUTPUT

```
1
1 2
1 2 3
1 2 3 4
```

18. Write a program for the use of goto statement

```
SET SERVEROUTPUT ON;
DECLARE
  n NUMBER := 1;
BEGIN
  <<my_loop>>
  DBMS_OUTPUT.PUT_LINE('Value of n: ' || n);
  n := n + 1;
  IF n <= 3 THEN
    GOTO my_loop;
  END IF;
  DBMS_OUTPUT.PUT_LINE('End of program');
END;
/
```

OUTPUT

```
Value of n: 1
Value of n: 2
Value of n: 3
End of program
```

19. Write a program to insert 10 row into square table

```
SET SERVEROUTPUT ON;
BEGIN
  FOR i IN 1..10 LOOP
    INSERT INTO square_table VALUES (i, i * i);
  END LOOP;
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('10 rows inserted successfully.');
```

END;
/

OUTPUT

10 rows inserted successfully.

20. Write a program to update row of square table

```
SET SERVEROUTPUT ON;
BEGIN
  FOR i IN 1..10 LOOP
    UPDATE square_table SET sq = i * i WHERE num = i;
  END LOOP;
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Rows updated successfully.');
```

END;
/

OUTPUT

Rows updated successfully.

21. Write a program to delete row from square table

```
SET SERVEROUTPUT ON;
BEGIN
  FOR i IN 1..5 LOOP
    DELETE FROM square_table WHERE num = i;
  END LOOP;
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Rows deleted successfully.');
```

END;
/

OUTPUT

Rows deleted successfully.